Lucas Callamullo – Full-Stack Software Engineer

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Profile

Experience designing modular architectures (microservices and monoliths). Stack across Java, .NET, React, automated testing and containerized deployments. More about me on my portfolio: **lucascallamullo.github.io**

Education

Universidad Tecnológica Nacional (UTN) — **Córdoba**, **ARG**
Information Systems Engineering — Mar 2023 – Present

Experience

Fleet Optimizer — Logistics Management System — Córdoba, ARG
Full-Stack Software Engineer | Personal Project — Jun 2026 – Sep 2026

- **Designed** a **microservices architecture** with 6 services in **Java** 17 and **Spring Boot** (auth, routes, packages, fleets, geocoding and gateway), with database-per-service where applicable.
- **Implemented** an **API Gateway** with **Spring Cloud** Gateway as the entry point, **handling JWT** validation and propagating user context (sub, roles, email) as headers to downstream microservices.
- Centralized **authentication and** role-based **authorization** with **Auth0** (**OAuth 2.0** / OIDC): token issuance, refresh and validation, with custom claims injected via Post-Login Actions.
- **Dockerized** each service with **multi-stage Dockerfiles** and orchestrated the full stack with **Docker Compose**, keeping **Nginx** as the only host-exposed entry point.
- Wrote **unit** and **integration tests** with **JUnit 5** and **Mockito**, covering success cases, errors and business rules.
- Built the **frontend** with **TypeScript**, **React** and **Tailwind** CSS (**responsive**), consuming the API through **Nginx**.
- Repository: github.com/LucasCallamullo/fleet-optimizer

E-commerce Engine — Córdoba, ARG
Full-Stack Software Engineer | Personal Project — Sep 2026 – Present

- **Designed** a **modular monolith** with **Onion architecture** in **C#** and **.NET**, with modules for Users, Products, Carts, Orders, Payments (**Mercado Pago API**) and AdminPanel over a centralized **PostgreSQL** database.
- **Improved** response times through **composite database indexes**, **Redis** caching for frequent queries, query optimization with **EF Core**, and DTO mapping with **Mapster**.
- Implemented **JWT authentication** and role-based **authorization**, with JWT session handling on the client side.
- **Dockerized** the services: **.NET** API, PostgreSQL, **Redis** and **Nginx** serving **React** as a **reverse proxy** for static assets and the single entry point.
- Built the **frontend** with **TypeScript**, **React** and **Tailwind** CSS (**responsive**), consuming the API through **Nginx**.
- Wrote **unit tests** with **xUnit**, covering success cases, errors and business rules.
- Repository: github.com/LucasCallamullo/ecommerce-engine

Technical Skills

- **Backend Java**: Java, Spring Boot, Spring Cloud, Spring Security.
- **Backend .NET:** C#, ASP.NET Core, EF Core.
- **Frontend:** React, TypeScript, Vite, Tailwind CSS.
- **Databases:** SQL, PostgreSQL, H2, Redis.
- **Testing & Tools:** JUnit 5, Mockito, xUnit, Swagger, Bruno, Ngrok.
- **Auth & Security:** OAuth 2.0, OpenID Connect (OIDC), JWT.
- **DevOps:** Docker, Docker Compose, Nginx, CI/CD, Git.

Languages

- **English:** B2 — fluent technical reading, written and spoken communication. | **Spanish:** Native.